

behaviours or biases. Data protection mandates that all training and inference processes adhere to applicable privacy laws and organisational policies. Employing techniques such as differential privacy, data anonymisation, and secure multi-party computation can mitigate risks associated with handling confidential information. Best practices also include minimising the retention of sensitive data, encrypting data at rest and in transit, and maintaining detailed access logs. By embedding these safeguards into the development lifecycle, organisations can foster trust and ensure ethical deployment of generative AI.

Handling hallucinations in open source LLMs

Hallucinations—outputs that are apparently fluent but false or not connected to reality—remain one of the most stubborn obstacles for open source LLMs to be used in production in IT, finance, healthcare, and legal processes. In the open source context, the challenge is becoming more formidable due to heterogeneity

in datasets, fine-tuning methodologies, and inference engine. Thus, mitigating hallucination necessitates a multi-layer solution from data, model alignment, retrieval grounding, decoding control and post-generation validation.

A good start is grounding. Retrieval-augmented generation (RAG) mitigates fabrication by making the model condition its answers on authoritative sources (policies, manuals, knowledge bases). Modern RAG implementations should not be limited to naive ‘top-k’ retrieval: use hybrid search (dense + BM25), rerankers (cross-encoders), document chunking fine-tuned to domain structure, and citation-aware prompting such that the response can be tied to retrieved passages. For business precision, add freshness controls (time-based filters) and access control aware retrieval so the model doesn’t cite content the user shouldn’t see.

Next, ‘align the model’s behaviour’. Instruction tuning and preference optimisation (e.g. DPO/IPO replacing classical RLHF pipelines) can diminish overconfident guessing by incentivising

behaviours like “I don’t know,” and “there is not enough evidence.” Where domain accuracy is important, employ domain-adaptive fine-tuning along with counterfactual/hard-negative training (penalises answers that are similar but wrong). It’s also useful to train with structured output constraints (JSON schema/function-calling style) so the model has to distinguish claims, evidence and uncertainty.

Hallucinations are also known to be correlated with sampling aggressiveness during inference time. Perform conservative decoding (decreasing temperature, using nucleus sampling bounds) and introduce logit biasing/refusal policies for unsafe or unverifiable questions. For tool-using agents, prefer tool-first patterns: the model ‘retrieves’, computes, or queries systems of record (CMDB, ticketing, inventory) rather than ‘memorising’ facts. Introduce self-consistency checks only when evidence-constrained (unbounded ‘chain-of-thought’ style deliberation can increase confident nonsense if retrieval is weak).

Table 2: A comparison of leading open source LLMs

Model	Key features	Advantages	Common use cases	Industry adoption
Llama (Meta)	Efficient transformer architecture, scalable sizes (7B to 65B parameters), multilingual capabilities	High performance on limited resources, flexible licensing, active community	Chatbots, research, content generation, code assistance	Technology, research, academia, startups
GPT-Neo/GPT-J (EleutherAI)	Open weights, transformer-based, compatible with GPT-3 benchmarks	Transparency, no usage restrictions, strong text generation	Automated writing, summarisation, language modelling	Media, education, research
Falcon	Optimised for inference speed and efficiency, open weights	Reduced operational costs, ease of deployment	Real-time applications, virtual assistants, customer service automation	Customer service, retail, SMEs
BLOOM	Multilingual, trained on diverse datasets, collaborative development	Wide language support, community-driven updates	Cross-lingual tasks, translation, social media analysis	Global enterprises, NGOs, research
MPT	Modular transformer, flexible context window, open licensing	Customisable, easy integration, suitable for long documents	Document processing, legal tech, knowledge management	Legal, compliance, enterprise